

Deterministic Optimization

Simplex Method in a Nutshell

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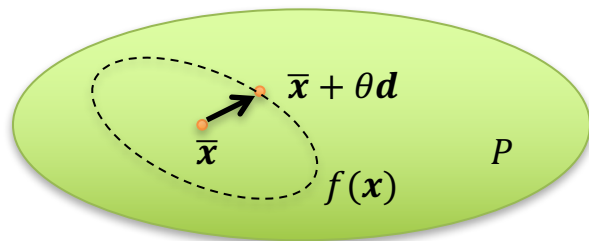
General Idea: Local Search

Simplex Method in a Nutshell

Learning Objectives for this module

- Identify the general algorithmic idea of local search
- Identify the geometric picture of applying local search to solving LP

A General Algorithmic Idea: Local Search



Algorithm 1 Local Search for minimizing $f(x)$ over $x \in P$

- 1: Start from a feasible solution $\bar{x} \in P$.
 - 2: Find a direction d that decreases the objective value.
 - 3: Find a step length θ so that the next iterate $\bar{x} \leftarrow \bar{x} + \theta d$ is feasible.
 - 4: If no such direction or step length can be found, terminate.
 - 5: $t \leftarrow t + 1$ and go to step 1.
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Gradient Descent & Newton Method

- Steepest descent:
 - Direction $\mathbf{d}$ is the negative gradient direction
 - $\mathbf{d} = -\nabla f(\bar{\mathbf{x}})$
- Newton's Method:
 - $\mathbf{d} = -[\nabla^2 f(\bar{\mathbf{x}})]^{-1} \nabla f(\bar{\mathbf{x}})$
- For constrained optimization, need to ensure feasibility
 - Need to ensure $\bar{\mathbf{x}} + \theta \mathbf{d}$ is still feasible

Local Search for Convex Optimization

- Can local search find an optimal solution?
- Yes, for convex optimization problems.
- Theorem: A local minimum of a convex program is also a global minimum.
- No, in general, for nonconvex optimization problems.
 - Local search can be stuck at a local minimum.

Summary

- We learned:
 - Local search as a generic algorithmic idea
 - It usually works for convex optimization problems, of course, when proper descent directions and step sizes are chosen